QF1100
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Theory of Interest

Principal Amount Sum of money for investment.
Interest Time value of money.
Accumulation Function Accumulated value of $\$1$ at time t, with $a(0) = 1$.
Simple Interest Linear Growth: $a(t) = 1 + rt$.
Compound Interest Exponential Growth: $a(t) = (1 + r)^t$.
Frequency of Compounding If an interest is paid p times a year, p is the frequency of compounding, $r^{(p)}$ is the nominal interest rate, and the interest paid each time is $\frac{r^{(p)}}{p}$.
Effective Annual Interest Rate $a(1) = 1 + r_e = (1 + \frac{r^{(p)}}{p})^p$, with accumulation function $a(t) = (1 + r_e)^t = (1 + \frac{r^{(p)}}{p})^{pt}$. The effective interest rate is always greater than the nominal rate.
Equivalent Interest Rates Two nominal interest rates are equivalent iff they yield the same accumulation amount over a year, i.e. $1 + r_e$ is equivalent.
Continuous Compounding For a positive interest rate, $lim_{x\to\infty}(1 + \frac{r}{x})^x = e^r$. Interest is compounded continuously when the frequency of compounding tends to infinity. The accumulation function is $a(t) = e^{r^{(\infty)}t}, t \geq 0$.
Force of Interest The force of interest, is $\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt}[\ln(a(t))]$.

- When $\delta(t) = \delta, a(t) = e^{\delta t}$, δ is the continuously compounded rate of interest.
- For simple interest, $\delta(t) = \frac{r}{1+rt}$.
- For compound interest, $\delta(t) = \ln(1 + r)$.

From definition, $a(s, t) = \exp(\int_s^t \delta(u)du)$. This implies the principle of consistency: $a(t_0, t_n) = a(t_0, t_1)a(t_1, t_2)...a(t_{n-1}, t_n)$.

Present Value

Present Value The present value of a payment c at time $t \geq 0$ is $\frac{c}{a(t)}$. Finding the present value of a future payment is known as discounting.
PV of Cash Flow Stream For a cash flow which is a series of payments, the present value $PV(C) = \sum_{i=1}^n \frac{c_i}{a(t_i)}$. If the effective annual interest rate is r, then $PV(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i}}$.
Time Value The time value of a cash flow $TV_t(C) = PV(C) \cdot a(t)$. For $0 < s < t, TV_t(C) = \frac{a(t)}{a(s)} \cdot TV_s(C)$. If the effective annual interest rate is r, then $TV_t(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i - t}}$ and $TV_t(C) = TV_s(C) \cdot (1 + r)^{t-s}$.
Principle of Equivalence Two cash flow streams are equivalent iff they have the same present value.
Net Present Value Considers PV of all negative and positive values, measure of value created today by undertaking investment.
Internal Rate of Return Non-negative root r of equation of value where $PV = 0$, a.k.a yield or IRR.
Unique IRR If unique, the IRR is the rate of return on investment. This occurs when $c_0 < 0, c_k \geq 0, c_0 + c_1 + ... + c_n > 0$. Otherwise, if the equation of value has degree n, then there are at most n distinct solutions.
Newton-Raphson There is no closed form solution for $n \geq 5$. First use IVT to pin down to a small interval where sign changes followed by choosing an initial guess. Then $\alpha_1 = \alpha_0 - \frac{f(\alpha_0)}{f'(\alpha_0)}$, repeat until satisfied.
Using IRR IRR as a measure of "survivability" can be used to compare investments, just like NPV, higher is better.

Annuities

Annuity Series of payments made at regular intervals, cash flow stream with evenly spread payments.
Ordinary Annuity Payments are made at the end of each period, a.k.a annuity immediate or annuity in arrears. $PV = \sum_{t=1}^n \frac{A}{(1+r)^t} = \frac{A}{r}[1 - \frac{1}{(1+r)^n}]$.
Annuity Due Payments are made at the beginning of each period. $PV = \sum_{t=0}^{n-1} \frac{A}{(1+r)^t} = \frac{A(1+r)}{r}[1 - \frac{1}{(1+r)^n}]$. $PV(OA) = \frac{1}{1+r}PV(AD)$.
Time Value of Annuities $TV_t = PV \cdot (1 + r)^t$.
Perpetual Annuity Pays a fixed sum forever. If Due, $PV = \sum_{t=0}^\infty \frac{A}{(1+r)^t} = \frac{A(1+r)}{r}$. If Ordinary, $PV = \sum_{t=1}^\infty \frac{A}{(1+r)^t} = \frac{A}{r}$.
Annuity Formula Cash flow of ordinary annuity is difference between two perpetuities, $PV = \frac{A}{r} - \frac{A}{r} \cdot \frac{1}{(1+r)^n} = \frac{A}{r}[1 - \frac{1}{(1+r)^n}]$

Loans

Loan Repaid by series of installment payments at periodic intervals. Cash flow is $+L, -A, -A, ...$.
Refinancing Revise terms of existing loan, based on balance amount. The balance amount is the time value of the remaining unpaid payments.
Prospective Method Consider cash flows for $t > m$, then $L_{Bal} = \frac{A}{1+r} + ... + \frac{A}{(1+r)^{n-m}} = \frac{A}{r}[1 - \frac{1}{(1+r)^{n-m}}]$.
Retrospective Method Consider cash flows for $t \leq m$, then $L_{Bal} = L(1 + r)^m - \frac{A}{r}[1 - \frac{1}{(1+r)^m}](1 + r)^m$.
Number of Payments Find number of payments, loan will be repaid with $\lfloor n \rfloor$ full payments of A then a last payment of $B < A$.

Bonds

Bond Written contract between issuer and investors

- **Face** / **Par Value** F, Amount based on which periodic interest payments are computed.
- **Redemption** / **Maturity Value** R, Amount to be repaid at the end of the loan.
- **Maturity** / **Redemption Data** Date on which loan will be fully repaid.
- **Coupon Rate** c, Bond's interest payments, as a percentage of par value, made to investors at regular intervals during term of loan.
- **Other Symbols** P: current price of a bond, m: number of coupon payments per year, n: total number of coupon payments, λ: nominal yield.

Law of One Price Price of bond equals to PV of cash flow consisting of all coupon payments and redemption value at maturity, calculated at yield λ.
Coupon Bonds Cash flow is made up of initial $-P$ and coupon payments of cF/m at multiples of $t = 1/m$, with redemption or maturity value $cF/m + R$ at time of last payment.
Valuing Coupon Bond
$P = \frac{R}{(1+\lambda/m)^n} + \sum_{i=1}^n \frac{cF/m}{(1+\lambda/m)^i} = \frac{R}{(1+\lambda/m)^n} + \frac{F \cdot c/m}{\lambda/m}(1 - \frac{1}{(1+\lambda/m)^n})$, made up of PV of redemption value and PV of coupon stream. Typically, $F = R$, then $P = F[\frac{1}{(1+\lambda/m)^n} + \frac{c/m}{\lambda/m}(1 - \frac{1}{(1+\lambda/m)^n})]$. Alternatively, $P = F + F(\frac{c-\lambda}{\lambda})[1 - \frac{1}{(1+\lambda/m)^n}]$.
Bond Purchase Premium: $P > F, c > \lambda$, Par: $P = F, c = \lambda$, Discount: $P < F, c < \lambda$.
Makeham Formula PV of face value at maturity is $K = \frac{F}{(1+\lambda/m)^n}$. $P = K + \frac{c}{\lambda}(F - K)$. Also, $P_{k+1} = P_k(1 + \lambda/m) - cF/m$, made up of accrued interest and coupon payment, after the kth payment.
Zero Coupon Bonds Do not pay coupons, cash flow is maturity value at the end. Then $P = \frac{R}{(1+\lambda)^N}$.
Perpetual Bond Never matures, $P = \frac{cF}{\lambda}$.
Price Yield Relationship P is a decreasing function of λ, and the curve is convex (bowl shaped).

Macaulay Duration Weighted average time to maturity of the cash flow from a bond. $D = \frac{\sum_{i=1}^n t_i PV\{(c_i, t_i)\}}{PV(C)} = \sum_{i=1}^n w_i t_i$. If $c_i \geq 0$, then $t_1 \leq D \leq t_n$. For a zero coupon bond with maturity T, $D = T$.
Yield Curve Relationship between yield and maturity

- Upward Sloping / Normal: Periods of economic expansion with low inflation.
- Downward Sloping / Inverted: Recessionary periods or when inflation is high.

Spot Rate Annual interest rate for money held from today until a future time, s_t.
Term Structure of Interest Rates Collection of spot rates.
Rate Conventions Assume a yearly convention, so that $(1 + s_t)^t$ is the factor by which a deposit held for t years will grow.
Spot Rate Curve Plot of spot rates against time to maturity. Yield curve can be determined from spot rates.
Forward Rate Interest rate observed at some future time. $f_{0,t} = s_t$, $(1 + s_k)^k = (1 + s_j)^j(1 + f_{j,k})^{k-j}$ for $k > j > 0$.
Determining Spot Rate Yield to maturity of a zero coupon bond that matures at t. Then $P = \frac{R}{(1+s_n)^n}$, $s_n = (\frac{R}{P})^{1/n} - 1$.
Bootstrap Method Suppose we have determined previous spot rates. Then we estimate s_n using the n-period bond with cash flow $(-P_n, cF/m, ..., cF/m + R)$. Using the pricing formula $P_n = \frac{cF/m}{1+s_1} + ... + \frac{cF/m+R}{(1+s_n)^n}$, we can find s_n. Use the zero-coupon bond to estimate s_1, 2 period coupon paying bond for s_2, etc.

Asset Returns

Asset Investment instrument that can be bought and sold.
Return on Investment $R = \frac{\text{amount received}}{\text{amount invested}} = \frac{P_1}{P_0}$ assuming it is purchased at P_0 and sold at P_1.
Rate of Return $r = \frac{P_1 - P_0}{P_0}$. Then we have $R = 1 + r, P_1 = (1 + r)P_0$, acting like a interest rate.
Short Sales Selling an asset that one does not own. Risky as potential loss is unlimited. Return is $R = \frac{-P_1}{-P_0} = \frac{P_1}{P_0}$. Net return $r = \frac{P_1}{P_0} - 1$.
Returns as Random Variables Money to be obtained when selling in the future is uncertain.
Random Variable Function that assigns a real number to each outcome in the sample space of a random experiment. It is discrete if its possible values are finite or countably infinite, if not it is continuous.
Expected Value $E(X) = \sum x \cdot p(x)$, $E(aX + bY) = aE(X) + bE(Y)$.
Variance $Var(X) = E[(X - E(X))^2] = E(X^2) - [E(X)^2] = \sigma^2$.
Standard Deviation $SD(X) = \sqrt{Var(X)} = \sigma$, has the same units as X.
Covariance Measure of linear dependence of two or more random variables. $\sigma_{X,Y} = Cov(X, Y) = E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y)$. Note $E(XY) \neq E(X)E(Y)$ in general. $Cov(X, X) = Var(X)$. $Cov(aX, bY) = abCov(X, Y)$. $Cov(aX + bY, Z) = aCov(X, Z) + bCov(Y, Z)$.
Correlation Coefficient $\rho_{X,Y} = Corr(X, Y) = \frac{Cov(X,Y)}{SD(X)SD(Y)}$, with absolute value less than 1. The magnitude measures the degree of linearity while the sign indicates the direction.
Independence $P(X = x, Y = y) = P(X = x)P(Y = y)$. Independence $\Rightarrow Cov = 0$.
Variance of Sum $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.

Portfolio Returns and Mean Variance Analysis

Mean and Variance of Return $\mu_i = E(r_i), \sigma_i^2 = Var(r_i)$. Mean measures return while SD measures risk.
Covariance of Returns Measure of linear dependence between returns of two assets $\sigma_{ij} = Cov(r_i, r_j)$. A standardised measure is the correlation coefficient $\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$.
Portfolio Splitting an initial amount X_0 to distribute as X_{0i} among n assets. It can be negative if short selling is allowed.
Portfolio Weights $X_{0i} = w_i X_0$. The sum must be equal to 1. It can be negative if short selling is allowed.
Portfolio of 2 Assets For initial wealth X_0, the wealth at the end of a period is $X_1 = X_0[1 + \alpha r_1 + (1-\alpha) r_2]$. The rate of return is $\alpha r_1 + (1-\alpha) r_2$.
Portfolio of n Assets $X_1 = X_0 \sum_{i=1}^n w_i(1+r_i) = X_0(1 + \sum_{i=1}^n w_i r_i)$. The rate of return is $X_1 = X_0(1 + r_p)$, then $r_p = \sum_{i=1}^n w_i r_i$. It is a linear combination of the asset returns.
Portfolio Mean and Variance $\mu_p = E(r_p) = \sum_{i=1}^n w_i \mu_i$.
$\sigma_p^2 = Var(r_p) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij}$ where $\sigma_{ij} = Cov(r_i, r_j)$.
Mean and Variance for Portfolio with 2 Assets $\mu_p = \alpha\mu_1 + (1-\alpha)\mu_2$.
$\sigma_p^2 = \alpha^2\sigma_1^2 + (1-\alpha)^2\sigma_2^2 + 2\alpha(1-\alpha)\rho_{12}\sigma_1\sigma_2$ using variance of sum.
Risk Aversion To reduce risk, seek optimal value of α that minimises σ_p^2.
Global Minimum Variance Portfolio $\sigma_p^2 =$
$(\alpha\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2} - \frac{\sigma_2(\sigma_2 - \rho_{12}\sigma_1)}{\sqrt{\sigma_1^2+\sigma_2^2-2\rho_{12}\sigma_1\sigma_2}})^2 + \sigma_2^2 - \frac{\sigma_2^2(\sigma_2-\rho_{12}\sigma_1)^2}{\sigma_1^2+\sigma_2^2-2\rho_{12}\sigma_1\sigma_2}$.
It occurs when $\alpha = \alpha^* = \frac{\sigma_2(\sigma_2-\rho_{12}\sigma_1)}{\sigma_1^2+\sigma_2^2-2\rho_{12}\sigma_1\sigma_2}$, with minimum portfolio variance $(\sigma_p^2)^* = \frac{\sigma_1^2\sigma_2^2(1-\rho_{12}^2)}{\sigma_1^2+\sigma_2^2-2\rho_{12}\sigma_1\sigma_2}$. The portfolio mean is then $(\mu_p)^* = \alpha^*\mu_1 + (1-\alpha^*)\mu_2$.
Feasible Sets Graph of μ_p against σ_p for all portfolio weights α. When $\rho = 1$, it is a straight line connecting $(\sigma_1, \mu_1), (\sigma_2, \mu_2)$. When $\rho = -1$, it is V-shaped graph connecting each (σ_i, μ_i) to a point with zero portfolio variance. Otherwise, it is a curve passing through $(\sigma_1, \mu_1), (\sigma_2, \mu_2)$.
Efficient Frontier Subset where $\mu_p \geq \mu_{GMVP}$

Forwards and Futures

Financial Derivative Financial instrument that has value derived from values of other more basic underlying assets (stocks, indices, commodities, interest rate, currencies).
Forward Contract
- Obligation to buy or sell an asset at a specific price and time in the future.
- t: Time where contract was agreed.
- T: Expiration date of forward contract.
- The buyer is long while the seller is short.
- $F(t,T)$: Forward price that applies at delivery.
- Spot Market: Open market for immediately delivery of asset, with spot price $S(t)$.
- Payoff to a position at time T is value of position at maturity. The profit subtracts the investment cost in the position at time T.
- For a long (will buy) forward contract, the payoff and profit is $S_T - F(t,T)$, while for a short (will sell) forward contract, it is $F(t,T) - S_T$.

Arbitrage Buying in one place and selling in another to make riskless profit from price differences.
No-Arbitrage Principle There cannot be two portfolio strategies with same initial cost such that one always has a better final payoff than the other.
Law of One Price If two portfolios have the same payoff, they must have the same value / initial price.
Forward Price By no-arbitrage, the forward price is $F(0,T) = S_0 e^{rT}, F(t,T) = S_t e^{r(T-t)}$.

Futures Contract

- Attempt to overcome liquidity and counterparty risk.
- Standardised legal contract to buy or sell something at a predetermined price for delivery at a specified time in the future between parties which are not yet know to each other.
- Margin Account: Indiviuals must open them with brokers, who require an initial margin amount as a fraction of contract value as a guarantee that contract holders will not default.
- If value of margin account drops below defined maintenance margin level, margin call is issued to contract holder; if not position will be closed.

Hedging

Hedging with Forward Contract Enter an agreement to remove the risk of price fluctuations in the spot market.
Perfect Hedge with Futures
- Risk associated with future commitment is completely eliminated by taking an opposite position in the futures market.
- Trader must match delivery date of asset to delivery date of futures.
- Commodity to be hedged must exactly match that of the futures contract.
- Amount of asset obligated must be integral multiple of contract size.
- There must be enough liquidity in the futures market.

Minimum variance Hedge
- If a perfect hedge for an asset is not available, one can manage risk using futures of another asset.
- Suppose you have W of A which you want to sell at T.
- Suppose you only have futures of B which is correlated to A.
- You can hedge by shorting h of B, where you want to minimise the risk of cashflow at T.
- Basis: Spot Price of asset to be hedged - Future Price of contract used.
- $S_A(T)$: Spot price of A at T.
- $F_B(0), F_B(T)$: Futures price of B at $t=0, t=T$.
- h: Future position taken.
- y_T: Cash flow at T.
- Cash Flow: Original Obligation + Profit in futures, $y = -WS_T + (F_T - F_0)h$.
- $Var(y) = W^2 Var(S_T) - 2WCov(S_T, F_T)h + Var(F_T)h^2$.
- To minimise $Var(y)$, set $-2WCov(S_T,F_T) + 2Var(F_T)h = 0$ (derivative wrt h).
- Then $h = W\frac{Cov(S_T,F_T)}{Var(F_T)} = W\frac{\rho_{S,F}\cdot\sigma_S}{\sigma_F}$,
$Var(y) = W^2Var(S_T) - \frac{W^2Cov(S_T,F_T)^2}{Var(F_T)} =$
$W^2Var(S_T)[1-\rho_{F,S}^2], \sigma_y = \sqrt{1-\rho_{F,S}^2} \times W\sigma_S$.
- $\beta = \frac{Cov(S_T,F_T)}{Var(F_T)}$ is the optimal hedge ratio.
- $\sqrt{1-\rho_{F,S}^2}$ measures effectiveness of hedging (most effective when 0).
- The SD $W\sigma_S$ of WS_T is the risk with no hedge.
- If sold instead, $h = -\beta W$.

Options

- Right but not obligation, to buy or sell an underlying asset under specified terms.
- Call / Put Option: Right to buy /sell specific quantity of asset for a strike price on or before a expiration date from the writer of the option.
- Holder of option pays writer option premium up front, and is in a long position.
- American options can be exercised anytime before the expiration date while European options (assumed) can only be exercised on the expiration date.

Option Value

- Call Option: For price S_T at expiration date, if $S_T < K$, the value is 0 by not exercising; if $S_T \geq K$,the value is $S_T - K$ by exercising it.
- Put Option: $K - S_T$ instead if it is positive.
- Long Position, Call Option: $\max(0, S_T - K) - TV_{t=T}(C)$
- Long Position, Put Option: $\max(0, K - S_T) - TV_{t=T}(P_0)$
- Short Position, Call Option: $-\max(0, S_T - K) + TV_{t=T}(C_0)$
- Short Position, Put Option: $-\max(0, K - S_T) + TV_{t=T}(P)$

Option Price Bounds

- Assumptions: No transaction costs; All trading profits are subject to the same tax rate; Borrowing and lending are possible at the same risk-free interest rate; Underlying assets are non-dividend-paying.
- Notation:
  - S_0: Current stock price; K: Strike price of option
  - T: Time to expiration of option; S_T: Stock price on expiration date
  - $r \geq 0$: Continuously compounded risk-free rate of interest
  - C_0: Price of European call option; P_0: Price of European put option
- Using the no-arbitrage assumption, we have $C_0 \leq S_0$ for calls and $P \leq Ke^{-rT}$ for puts (discounted strike price).
- For calls, we have $\max(0, S_0 - Ke^{-rT}) \leq C$, from $S_0 \leq C + Ke^{-rT}$.
- For puts, we have $\max(0, Ke^{-rT} - S_0) \leq P$, from $Ke^{-rT} \leq P + S_0$.
- Arbitrage Strategies:
  - $C_0 < S_0 - Ke^{-rT}$: Buy Call, Short Stock, Lend Cash (Invest $S_0 - C$).
  - $P_0 < Ke^{-rT} - S_0$: Buy Put, Buy Stock, Borrow Cash (Borrow $S_0 + P$).
  - $C + Ke^{-rT} < P + S_0$ (Parity): Buy Call, Borrow, Short Put, Short Stock. $C + Ke^{-rT} > P + S_0$: Short Call, Invest, Long Put, Long Stock.

Trading Strategies

- Principal-protected notes: Created from zero-coupon bond and a European call option. Issuer guarantees that purchaser will receive principal back regardless of performance of asset underlying the option.
- Common strategy involves single option and underlying stock.
- Spread: Take a position in multiple options of the same type.
- Combination: Take a position in both calls and puts on the same stock.
- Butterfly Spread: Consider $K_1 < K_2 < K_3$. Long a call with K_1, K_3. Short calls with price K_2. Payoff is positive between K_1, K_3, peaking at K_2.
- Bull Spread: Long call K_1, short call K_2, positive payoff when higher than K_1.
- Covered Calls: Long position in underlying asset and short position in call option on same asset.
- Protective Puts: Long position in underlying asset and long position in a put option on the same asset.
- Long Straddles: Long positions in a European call and put with same maturity and strike price.

Put-Call Parity

- Relation between prices of European put and call options with same strike price and time to maturity.
- C_0, P_0: Price of European call and put
- K: Strike price; S_0: Current stock price
- d: Discount factor to maturity (expiration date)
- Put-call parity states that $C + dK = P + S_0$, for no-arbitrage.

Binomial Model for Option Pricing

- S: Initial price of stock; $u > d > 0$: Up / down factors; r: Risk free interest rate
- At the end of one period, price will be uS with probability p and price will be dS with probability $1-p$. With no-arbitrage, we must have $u > R = 1 + r > d$.
- The value of a call option is either $(uS - K)^+$ or $(dS - K)^+$.
- Let $C_u = \max(uS - K, 0), C_d = \max(dS - K, 0)$.
- By no-arbitrage, $C_0 = \frac{1}{R}[qC_u + (1-q)C_d]$, where $q = \frac{R-d}{u-d}$ is the risk-neutral probability. This is independent of p.
- Replicating portfolio: (x, b) of (stock, cash), $x = \frac{C_u - C_d}{u-d}, b = \frac{uC_d - dC_u}{R(u-d)}$.